



10246 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
CenA-Inner				

JWST Proposal 10246 (Created: Wednesday, May 13, 2026, 11:01:49AM Eastern Standard Time) - Overview

Folder	Observation	Label	Observing Template	Science Target
	1	CenA-Inner-Pre-Imaging	NIRCam Imaging	(11) CenA-Inner-MSA-Catalog-Mock
	2	CenA-Inner-PA215-2nod-mock	NIRSpec MultiObject Spectroscopy	(11) CenA-Inner-MSA-Catalog-Mock
CenA-Middle				
	3	CenA-Middle-Pre-Imaging	NIRCam Imaging	(8) CenA-Middle-MSA-Catalog-Mock
	4	CenA-Middle_PA215-2nod-mock	NIRSpec MultiObject Spectroscopy	(8) CenA-Middle-MSA-Catalog-Mock
CenA-Stream				
	5	CenA-Stream-Pre-Imaging	NIRCam Imaging	(9) CenA-Stream-MSA-Catalog-Mock
	6	CenA-Stream_PA70-2nod-mock	NIRSpec MultiObject Spectroscopy	(9) CenA-Stream-MSA-Catalog-Mock
CenA-Outer				
	8	CenA-Outer-Pre-Imaging	NIRCam Imaging	(10) CenA-Outer-MSA-Catalog-Mock
	9	CenA-Outer_PA260-2nod-mock	NIRSpec MultiObject Spectroscopy	(10) CenA-Outer-MSA-Catalog-Mock

ABSTRACT

We propose to leverage JWST's unique spectroscopic and imaging capabilities in Centaurus A (Cen A; 3.8 Mpc) to perform the first resolved star chemodynamical study of a massive early type galaxy (ETG). Using NIRSpec MSA spectroscopy, we will measure stellar [Fe/H] and [alpha/Fe] to a precision of 0.1–0.3 dex and radial velocities to ~5 km/s for ~300 RGB stars across 4 fields located along Cen A's semi-major axis (25-85 kpc; 5-17 effective radii), including in its most prominent substructure, the CenA-MM-Dw3 stellar stream. In parallel, we will acquire deep NIRCam photometry extending 1 magnitude below the red clump. With these observations, we will produce the first distributions and radial profile of stellar [Fe/H] and [alpha/Fe] measurements in any giant elliptical galaxy, the most detailed star formation history measurement in an ETG to date, new constraints on Cen A's dark matter halo mass and shape, and the first comprehensive characterization of Cen A's population of asymptotic giant branch stars. This program will provide transformative new results on the chemical enrichment, mass assembly, and formation of massive ETGs. Only JWST has the wavelength coverage, sensitivity, and angular resolution to make these measurements in the distant and crowded halo of Cen A—no other ground- or space-based facility can acquire this data.

OBSERVING DESCRIPTION

In this program, we will obtain NIRSpec MOS of ~ 300 tip of the red giant branch (TRGB) and ~ 20 asymptotic giant branch (AGB) stars and deep NIRCам imaging in 4 fields of the Cen A halo. These fields will be placed along Cen A's semi-major axis located at galactocentric distances of ~ 25 , 50, 75, and 85 kpc (5, 10, 15, and 17 effective radii, respectively). The third and fourth pointings are placed such they collectively observe with both NIRSpec and NIRCам the CenA-MM-Dw3 stream's northeast arm as well as the adjacent smooth halo.

Targets for NIRSpec MSA will be selected from NIRCам pre-imaging in the F150W and F227W filters, which we will acquire using the FULLBOX 8NIRSPEC dither and BRIGHT1 readout patterns and a total of 2 dithers, 1 integration per dither, and 5 groups per integration. We fill our NIRSpec MSA masks preferentially with stars within 1 mag of the TRGB ($24.9 < F814W < 23.9$) for which $[Fe/H]$ and $[\alpha/Fe]$ can be measured to 0.1–0.3 dex precision (~ 50 -100 stars/field). We will then assign available slits to AGB stars just above the TRGB ($23.9 < F814W < 23$) for which we can also perform AGB classification (5-10 stars/field). We use 1-shutter slitlets and adopt the tightest (< 55 mas) slit centering constraint when assigning stars to slits in order to minimize slit losses and radial velocity systematics. The remainder of the mask will be filled with ~ 40 "blank sky" shutters where we know no stars with $F814W < 26$ exist in order to determine and correct for the unresolved background spectrum. The stellar density of these fields is high enough to efficiently utilize the multiplexing of NIRSpec's MSA, but not so high as to over-crowd MSA slitlets. Furthermore, this field has many suitable stars for MSATA without violating crowding constraints.

NIRSpec observations will be taken with the G140H/F100LP disperser-filter pair, providing 0.97-1.82 micron spectral coverage at an effective resolving power of $R \sim 2700$. At this modest resolution, measuring $[\alpha/Fe]$ and $[Fe/H]$ to 0.1-0.3 dex precision can be achieved in our sample with spectra of $S/N > 15$ per pixel at 1.4 micron. This requires ~ 23 hours of integration for our faintest RGB stars ($F814W = 24.9$). We will take 6 exposures split across a 2-point 1-shutter nod in both the dispersion and cross-dispersion directions (to mitigate detector effects and cosmic rays) and a sub-pixel dither in the dispersion direction (to improve spectral sampling). Each NIRSpec exposures will employ the NRSIRS2 readout pattern with 9 integration of 21 groups. We request confirmation imaging for each slit mask configuration to quantify slit losses and improve velocity extraction. The median S/N of our TRGB sample will be ~ 20 per pixel at 1.4 micron, while spectra for the AGB stars will have a S/N as high as ~ 50 per pixel. In addition to stellar chemical abundances, we will measure radial velocities for all stars to ~ 5 km/s precision.

NIRCам coordinated parallels will image the Cen A halo in adjacent fields in two SW filters (F090W, F150W) to construct a deep CMD extending below the red clump ($F814W \sim 29.5$) and 3 LW filters (F250M, F430M, F444W) to characterize the Cen A population of AGB stars. We will take a total of 6 sets of NIRCам exposures (set by the 2-point nod and 3-point sub-pixel dither pattern of NIRSpec) using the MEDIUM8 readout and

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3POINT-WITH-NIRCam-SIZE1 dither patterns. Each set of exposures consists of 3 dithers, 8 integrations/dither, and 5 groups/integration. To achieve the required S/N in the SW channel, we will split the exposure sets 2:1 between the F090W and F150W filters. In the LW channel, the required S/N will be achieved by allocating 1, 4, and 1 exposures to the F250M, F430M, and F444W filters respectively.

Proposal 10246 - Targets - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(8)	CenA-Middle-MSA-Catalog-Mock	RA: 13 27 31.9833 (201.8832637d) Dec: -42 12 25.76 (-42.20716d) Equinox: J2000		
	Comments: Description=[]				
	(9)	CenA-Stream-MSA-Catalog-Mock	RA: 13 28 29.9679 (202.1248663d) Dec: -41 51 2.65 (-41.85074d) Equinox: J2000		
	Comments: Description=[]				
	(10)	CenA-Outer-MSA-Catalog-Mock	RA: 13 28 29.9680 (202.1248667d) Dec: -41 42 2.65 (-41.70074d) Equinox: J2000		
	Comments: Description=[]				
	(11)	CenA-Inner-MSA-Catalog-Mock	RA: 13 26 28.9824 (201.6207600d) Dec: -42 38 31.47 (-42.64207d) Equinox: J2000		
	Comments: Description=[]				

Proposal 10246 - Observation 1 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Observation	Proposal 10246, Observation 1: CenA-Inner-Pre-Imaging Wed May 13 16:01:49 GMT 2026										
	Diagnostic Status: Warning Observing Template: NIRCcam Imaging										
Diagnostics	(CenA-Inner-Pre-Imaging (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.										
	(CenA-Inner-Pre-Imaging (Obs 1)) Warning (Form): Some science pointings place the target outside of the aperture.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(11)	CenA-Inner-MSA-Catalog-Mock	RA: 13 26 28.9824 (201.6207600d) Dec: -42 38 31.47 (-42.64207d) Equinox: J2000								
	<i>Comments:</i> Description=[]										
Template	Module		Subarray			Target Placement					
	ALL		FULL			Module gap (large extended source)					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	FULLBOX		8NIRSPEC		STANDARD				3	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F277W	BRIGHT1	5	1	24	24	2319.142		
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 69.92542306 to 124.92542306 Degrees (V3 70.0 to 125.0) Aperture PA Range 284.92542306 to 339.92542306 Degrees (V3 285.0 to 340.0) Visits Same PA										
	2 After 1 by 60.0 Days to <None specified>										

Proposal 10246 - Observation 2 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Observation	Proposal 10246, Observation 2: CenA-Inner-PA215-2nod-mock											Wed May 13 16:01:49 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCam Imaging											
Diagnostics	(CenA-Inner-PA215-2nod-mock (Obs 2)) Warning (Form): Config c1 (#1) has 1 primary slits affected by failed closed shutters.											
	(CenA-Inner-PA215-2nod-mock (Obs 2)) Warning (Form): Config c1 (#2) has 1 primary slits affected by failed closed shutters.											
	(CenA-Inner-PA215-2nod-mock (Obs 2)) Warning (Form): Config c1 (#3) has 1 primary slits affected by failed closed shutters.											
	(Visit 2:1) Warning (Form): Data Excess over lower threshold											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 2:2) Warning (Form): Data Excess over lower threshold											
	(Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(11)	CenA-Inner-MSA-Catalog-Mock	RA: 13 26 28.9824 (201.6207600d) Dec: -42 38 31.47 (-42.64207d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy						NIRCam Imaging					
	TA Method: MSATA						Module: ALL					
	HFF Readout Mode: false						Subarray: FULL					
	Obtain Confirmation Images: After Target ACQ and New MSA Config											
	Science Aperture: MSA Center											
	Primary Candidate List: CenA-Inner-TRGB-Mock (49383 sources)											
Reference Stars	Filler Candidate List: CenA-Inner-AGB-Mock (4590 sources)											
	Spectral Overlap Map: jwst-nirspec-g140h											
	Spectral Overlap Threshold: 1.5											
Dithers	#	Dither Type										
	1	3-POINT-WITH-NIRCam-SIZE1										

Proposal 10246 - Observation 2 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Confirmation	NIRSpec MultiObject Spectroscopy										
	Confirmation Type		Conf. Readout Pattern		Conf. Groups/Int		Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time
	1	c1	NRSIRS2RAPID		2		1		3		131.3
2	c2	NRSIRS2RAPID		2		1		3		131.3	
Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140H/F100LP)	c1		201.61985616666 666 Degrees - 42.64095333333 36 Degrees	215.00061528525 134	-0.1		3	9	13917.801
	2	1 (G140H/F100LP)	c1		201.61985616666 666 Degrees - 42.64095333333 36 Degrees	215.00060969428 205	0.0		3	9	13917.801
	3	1 (G140H/F100LP)	c1		201.61985616666 666 Degrees - 42.64095333333 36 Degrees	215.00060410330 772	0.1		3	9	13917.801
	4	1 (G140H/F100LP)	c2		201.61982429166 667 Degrees - 42.64111638888 92 Degrees	215.00063719652 04	-0.1		3	9	13917.801
	5	1 (G140H/F100LP)	c2		201.61982429166 667 Degrees - 42.64111638888 92 Degrees	215.00063160552 054	0.0		3	9	13917.801
	6	1 (G140H/F100LP)	c2		201.61982429166 667 Degrees - 42.64111638888 92 Degrees	215.00062601451 577	0.1		3	9	13917.801
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	2	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	3	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	4	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	5	F150W	F250M	MEDIUM8	5	8	24	3	12594.231		
	6	F150W	F444W	MEDIUM8	5	8	24	3	12594.231		
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 63.5745697 to 118.5745697 Degrees (V3 285.0 to 340.0) Aperture PA Range 208.5745697 to 263.5745697 Degrees (V3 70.0 to 125.0) Visits Same PA No Parallel Attachments MSA Planned Aperture PA 215.0000 to 215.0000 Degrees (V3 76.4254303 to 76.4254303)										
	2 After 1 by 60.0 Days to <None specified>										

Proposal 10246 - Observation 3 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Observation	Proposal 10246, Observation 3: CenA-Middle-Pre-Imaging									Wed May 13 16:01:49 GMT 2026
	Diagnostic Status: Warning									
Observing Template: NIRCam Imaging										
Diagnostics	(CenA-Middle-Pre-Imaging (Obs 3)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.									
	(CenA-Middle-Pre-Imaging (Obs 3)) Warning (Form): Some science pointings place the target outside of the aperture.									
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Visit 3:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(8)	CenA-Middle-MSA-Catalog-Mock	RA: 13 27 31.9833 (201.8832637d) Dec: -42 12 25.76 (-42.20716d) Equinox: J2000							
Comments: Description=[]										
Template	Module		Subarray				Target Placement			
	ALL		FULL				Module gap (large extended source)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	FULLBOX		8NIRSPEC		STANDARD				3
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F150W	F277W	BRIGHT1	5	1	24	24	2319.142	
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 69.92542306 to 124.92542306 Degrees (V3 70.0 to 125.0) Aperture PA Range 284.92542306 to 339.92542306 Degrees (V3 285.0 to 340.0) Visits Same PA									
	4 After 3 by 60.0 Days to <None specified>									

Proposal 10246 - Observation 4 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Observation	Proposal 10246, Observation 4: CenA-Middle_PA215-2nod-mock										Wed May 13 16:01:49 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Coordinated Parallel Template(s): NIRCcam Imaging										
Diagnostics	(CenA-Middle_PA215-2nod-mock (Obs 4)) Warning (Form): Config c1 (#1) has 1 primary slits affected by failed closed shutters.										
	(CenA-Middle_PA215-2nod-mock (Obs 4)) Warning (Form): Config c1 (#2) has 1 primary slits affected by failed closed shutters.										
	(CenA-Middle_PA215-2nod-mock (Obs 4)) Warning (Form): Config c1 (#3) has 1 primary slits affected by failed closed shutters.										
	(CenA-Middle_PA215-2nod-mock (Obs 4)) Warning (Form): Config c2 (#4) has 1 primary slits affected by failed closed shutters.										
	(CenA-Middle_PA215-2nod-mock (Obs 4)) Warning (Form): Config c2 (#5) has 1 primary slits affected by failed closed shutters.										
	(CenA-Middle_PA215-2nod-mock (Obs 4)) Warning (Form): Config c2 (#6) has 1 primary slits affected by failed closed shutters.										
	(Visit 4:1) Warning (Form): Data Excess over lower threshold										
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	(Visit 4:2) Warning (Form): Data Excess over lower threshold										
	(Visit 4:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	#	Name		Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(8)	CenA-Middle-MSA-Catalog-Mock		RA: 13 27 31.9833 (201.8832637d) Dec: -42 12 25.76 (-42.20716d) Equinox: J2000							
Acquisition	Comments: Description=[]										
	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	NIRSpec MultiObject Spectroscopy										
	NIRCcam Imaging										
	TA Method: MSATA										
	Module: ALL										
	HFF Readout Mode: false										
	Subarray: FULL										
	Obtain Confirmation Images: After Target ACQ and New MSA Config										
	Science Aperture: MSA Center										
Primary Candidate List: CenA-Middle-TRGB-Mock (2295 sources)											
Filler Candidate List: CenA-Middle-AGB-Mock (819 sources)											
Spectral Overlap Map: jwst-nirspec-g140h											
Spectral Overlap Threshold: 1.5											
Reference Stars											

Proposal 10246 - Observation 4 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Dithers	Dither Type										
	#										
	3-POINT-WITH-NIRCam-SIZE1										
Confirmation	NIRSpec MultiObject Spectroscopy		Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time			
	1		c1	NRSIRS2RAPID	2	1	3	131.3			
	2		c2	NRSIRS2RAPID	2	1	3	131.3			
Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140H/F100LP)	c1		201.88292779166 667 Degrees - 42.206659722222 24 Degrees	215.00022997076 3	-0.1		3	9	13917.801
	2	1 (G140H/F100LP)	c1		201.88292779166 667 Degrees - 42.206659722222 24 Degrees	215.00022446440 03	0.0		3	9	13917.801
	3	1 (G140H/F100LP)	c1		201.88292779166 667 Degrees - 42.206659722222 24 Degrees	215.00021895803 27	0.1		3	9	13917.801
	4	1 (G140H/F100LP)	c2		201.88289616666 665 Degrees - 42.206822777777 8 Degrees	215.00025153498 805	-0.1		3	9	13917.801
	5	1 (G140H/F100LP)	c2		201.88289616666 665 Degrees - 42.206822777777 8 Degrees	215.00024602859 534	0.0		3	9	13917.801
	6	1 (G140H/F100LP)	c2		201.88289616666 665 Degrees - 42.206822777777 8 Degrees	215.00024052219 757	0.1		3	9	13917.801
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	2	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	3	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	4	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	5	F150W	F250M	MEDIUM8	5	8	24	3	12594.231		
	6	F150W	F444W	MEDIUM8	5	8	24	3	12594.231		

Proposal 10246 - Observation 4 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 63.5745697 to 118.5745697 Degrees (V3 285.0 to 340.0) Aperture PA Range 208.5745697 to 263.5745697 Degrees (V3 70.0 to 125.0) Visits Same PA No Parallel Attachments MSA Planned Aperture PA 215.0000 to 215.0000 Degrees (V3 76.4254303 to 76.4254303) 4 After 3 by 60.0 Days to <None specified>
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Proposal 10246 - Observation 5 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Observation	Proposal 10246, Observation 5: CenA-Stream-Pre-Imaging										Wed May 13 16:01:49 GMT 2026
	Diagnostic Status: Warning Observing Template: NIRCam Imaging										
Diagnostics	(CenA-Stream-Pre-Imaging (Obs 5)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.										
	(CenA-Stream-Pre-Imaging (Obs 5)) Warning (Form): Some science pointings place the target outside of the aperture.										
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 5:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(9)	CenA-Stream-MSA-Catalog-Mock	RA: 13 28 29.9679 (202.1248663d) Dec: -41 51 2.65 (-41.85074d) Equinox: J2000								
Template	Comments: Description=[]										
	Module	Subarray			Target Placement						
	ALL	FULL			Module gap (large extended source)						
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	FULLBOX		8NIRSPEC		STANDARD				3	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F277W	BRIGHT1	5	1	24	24	2319.142		
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 69.92542306 to 124.92542306 Degrees (V3 70.0 to 125.0) Aperture PA Range 284.92542306 to 339.92542306 Degrees (V3 285.0 to 340.0) Visits Same PA										
	6 After 5 by 60.0 Days to <None specified>										

Proposal 10246 - Observation 6 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Observation	Proposal 10246, Observation 6: CenA-Stream_PA70-2nod-mock											Wed May 13 16:01:49 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCcam Imaging											
Diagnostics	(Visit 6:1) Warning (Form): Data Excess over lower threshold											
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 6:2) Warning (Form): Data Excess over lower threshold											
	(Visit 6:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 6:1) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
Fixed Targets	(Visit 6:2) Informational (Form): Visit schedulable, but most scheduling windows are when JWST is pointed in direction of greatest micrometeoroid impact risk. This is likely due to scheduling special requirements.											
	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(9)	CenA-Stream-MSA-Catalog-Mock	RA: 13 28 29.9679 (202.1248663d) Dec: -41 51 2.65 (-41.85074d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy											NIRCcam Imaging
	TA Method: MSATA											Module: ALL
	HFF Readout Mode: false											Subarray: FULL
	Obtain Confirmation Images: After Target ACQ and New MSA Config											
	Science Aperture: MSA Center											
	Primary Candidate List: CenA-Stream-TRGB-Mock (2376 sources)											
	Filler Candidate List: CenA-Stream-AGB-Mock (810 sources)											
Reference Stars	Spectral Overlap Map: jwst-nirspec-g140h											
	Spectral Overlap Threshold: 1.5											
Dithers	#	Dither Type										
	1	3-POINT-WITH-NIRCcam-SIZE1										

Proposal 10246 - Observation 6 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Confirmation	NIRSpec MultiObject Spectroscopy										
	Confirmation Type		Conf. Readout Pattern		Conf. Groups/Int		Conf. Integrations/Exp		Conf. Total Integrations		Conf. Total Exposure Time
	1	c1	NRSIRS2RAPID		2		1		3		131.3
2	c2	NRSIRS2RAPID		2		1		3		131.3	
Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140H/F100LP)	c1		202.12345116666 668 Degrees - 41.84877944444 46 Degrees	70.000944885592 16	-0.1		3	9	13917.801
	2	1 (G140H/F100LP)	c1		202.12345116666 668 Degrees - 41.84877944444 46 Degrees	70.000947169864 69	0.0		3	9	13917.801
	3	1 (G140H/F100LP)	c1		202.12345116666 668 Degrees - 41.84877944444 46 Degrees	70.000949454147 08	0.1		3	9	13917.801
	4	1 (G140H/F100LP)	c2		202.12360245833 332 Degrees - 41.84865916666 664 Degrees	70.000844265361 25	-0.1		3	9	13917.801
	5	1 (G140H/F100LP)	c2		202.12360245833 332 Degrees - 41.84865916666 664 Degrees	70.000846549635 17	0.0		3	9	13917.801
	6	1 (G140H/F100LP)	c2		202.12360245833 332 Degrees - 41.84865916666 664 Degrees	70.000848833918 92	0.1		3	9	13917.801
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	2	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	3	F150W	F430M	MEDIUM8	5	8	24	3	12594.231		
	4	F150W	F430M	MEDIUM8	5	8	24	3	12594.231		
	5	F150W	F250M	MEDIUM8	5	8	24	3	12594.231		
	6	F150W	F444W	MEDIUM8	5	8	24	3	12594.231		
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 70 to 100 Degrees (V3 291.4254303 to 321.4254303) Visits Same PA No Parallel Attachments MSA Planned Aperture PA 70.0000 to 70.0000 Degrees (V3 291.4254303 to 291.4254303)										
	6 After 5 by 60.0 Days to <None specified>										

Proposal 10246 - Observation 8 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Observation	Proposal 10246, Observation 8: CenA-Outter-Pre-Imaging										Wed May 13 16:01:49 GMT 2026
	Diagnostic Status: Warning										
Observing Template: NIRCcam Imaging											
Diagnostics	(CenA-Outter-Pre-Imaging (Obs 8)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.										
	(CenA-Outter-Pre-Imaging (Obs 8)) Warning (Form): Some science pointings place the target outside of the aperture.										
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 8:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(10)	CenA-Outter-MSA-Catalog-Mock	RA: 13 28 29.9680 (202.1248667d) Dec: -41 42 2.65 (-41.70074d) Equinox: J2000								
Comments: Description=[]											
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module gap (large extended source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	FULLBOX		8NIRSPEC		STANDARD				3	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F277W	BRIGHT1	5	1	24	24	2319.142		
Special Requirements	Sequence Visits within 53.0 Days Aperture PA Range 69.92542306 to 124.92542306 Degrees (V3 70.0 to 125.0) Aperture PA Range 284.92542306 to 339.92542306 Degrees (V3 285.0 to 340.0) Visits Same PA										
	9 After 8 by 60.0 Days to <None specified>										

Proposal 10246 - Observation 9 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Observation	Proposal 10246, Observation 9: CenA-Outter_PA260-2nod-mock											Wed May 13 16:01:49 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCcam Imaging											
Diagnostics	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c1 (#1) has 1 filler slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c1 (#1) has 2 primary slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c1 (#2) has 1 filler slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c1 (#2) has 2 primary slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c1 (#3) has 1 filler slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c1 (#3) has 2 primary slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c2 (#4) has 1 filler slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c2 (#4) has 1 primary slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c2 (#5) has 1 filler slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c2 (#5) has 1 primary slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c2 (#6) has 1 filler slits affected by failed closed shutters.											
	(CenA-Outter_PA260-2nod-mock (Obs 9)) Warning (Form): Config c2 (#6) has 1 primary slits affected by failed closed shutters.											
	(Visit 9:1) Warning (Form): Data Excess over lower threshold											
	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 9:2) Warning (Form): Data Excess over lower threshold											
(Visit 9:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(10)	CenA-Outter-MSA-Catalog-Mock	RA: 13 28 29.9680 (202.1248667d) Dec: -41 42 2.65 (-41.70074d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy						NIRCcam Imaging					
	TA Method: MSATA						Module: ALL					
	HFF Readout Mode: false						Subarray: FULL					
	Obtain Confirmation Images: After Target ACQ and New MSA Config											
	Science Aperture: MSA Center											
	Primary Candidate List: CenA-Outter-TRGB-Mock (2376 sources)											
	Filler Candidate List: CenA-Outter-AGB-Mock (810 sources)											
	Spectral Overlap Map: jwst-nirspec-g140h											
	Spectral Overlap Threshold: 1.5											

Proposal 10246 - Observation 9 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Reference Stars											
Dithers	#	Dither Type									
	1	3-POINT-WITH-NIRCam-SIZE1									
Confirmation	NIRSpec MultiObject Spectroscopy	Confirmation Type	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time				
	1	c1	NRSIRS2RAPID	2	1	3	131.3				
	2	c2	NRSIRS2RAPID	2	1	3	131.3				
Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G140H/F100LP)	c1		202.12693008333 335 Degrees - 41.698210277777 775 Degrees	259.99862233827 804	-0.1		3	9	13917.801
	2	1 (G140H/F100LP)	c1		202.12693008333 335 Degrees - 41.698210277777 775 Degrees	259.99862120355 35	0.0		3	9	13917.801
	3	1 (G140H/F100LP)	c1		202.12693008333 335 Degrees - 41.698210277777 775 Degrees	259.99862006881 983	0.1		3	9	13917.801
	4	1 (G140H/F100LP)	c2		202.1267535 Degrees - 41.698309166666 66 Degrees	259.99874012395 99	-0.1		3	9	13917.801
	5	1 (G140H/F100LP)	c2		202.1267535 Degrees - 41.698309166666 66 Degrees	259.99873898924 477	0.0		3	9	13917.801
	6	1 (G140H/F100LP)	c2		202.1267535 Degrees - 41.698309166666 66 Degrees	259.99873785452 06	0.1		3	9	13917.801
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	2	F090W	F430M	MEDIUM8	5	8	24	3	12594.231		
	3	F150W	F430M	MEDIUM8	5	8	24	3	12594.231		
	4	F150W	F430M	MEDIUM8	5	8	24	3	12594.231		
	5	F150W	F250M	MEDIUM8	5	8	24	3	12594.231		
	6	F150W	F444W	MEDIUM8	5	8	24	3	12594.231		

Proposal 10246 - Observation 9 - Exploring the Accretion History and Halo of Giant Elliptical Cen A with Resolved Stars

Special Requirements

Sequence Visits within 53.0 Days
Aperture PA Range 230 to 260 Degrees (V3 91.4254303 to 121.4254303)
Visits Same PA
No Parallel Attachments
MSA Planned Aperture PA 260.0000 to 260.0000 Degrees (V3 121.4254303 to 121.4254303)
9 After 8 by 60.0 Days to <None specified>